Ogbonna Arinze

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SUMMARY

Product-minded Software Engineer with 8 years of professional experience building full-stack applications from database to React UI across healthcare, fintech, civic tech, and enterprise environments. Demonstrated expertise in AI-native development, including production multi-agent systems with 5 specialized agents, hybrid RAG pipelines processing 708K+ vector embeddings, and agentic tool orchestration with Claude Code. Proven ability to translate business requirements into technical solutions, own features end-to-end in React, TypeScript, Next.js, NestJS, and PostgreSQL, and collaborate directly with technical leadership in fast-paced, early-stage environments.

CORE COMPETENCIES

- **AI & Agentic Development:** Multi-agent AI systems (Mastra), RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), embeddings, agentic tool orchestration, and Claude Code.
- **Fullstack Development:** React, Next.js, TypeScript, NestJS, NodeJS REST APIs, PostgreSQL, MongoDB, Prisma, and database-to-UI feature ownership.
- **Product Engineering:** End-to-end feature ownership, business requirement translation, scalable architecture design, and user-focused development.
- **Frontend Architecture:** Design systems (Radix UI), state management (Redux, Zustand), SSR/SSG, responsive layouts, and performance optimization.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Cloud & Infrastructure:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS, Nx monorepo orchestration, and Traefik.
- **Programming Languages:** TypeScript, JavaScript (ESNext), NodeJS, and Python.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources, embed the widget on external sites, and query their data in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, owning the migration from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing adoption.
- Implemented **JWT-based authentication** and **role-based access control** with HTTP-only cookies, collaborating closely with backend engineers to align on auth strategies and enforce permission-based UI rendering.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios**.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Translated business requirements into efficient **MongoDB** database models and schemas, developing **NodeJS RESTful APIs** focused on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. Multi-agent LLM pipeline analyzes 700+ budget documents (708K+ vector embeddings) across 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with 22 interactive chart types (Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, owning every feature from **PostgreSQL** database design to **React** UI.
- Built a **multi-agent AI system** using **Mastra** with **5 specialized agents** (Router, Budget Analyst, Corruption Analyst, GovSpend Analyst, Impact Analyst) and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback), powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, custom migration tooling, and a document ingestion pipeline supporting PDF (OCR via Tesseract), XLSX, DOCX, JSON, and Markdown, processing 700+ documents with zero ingestion errors.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management, conversation summarization, and memory extraction. Built phone OTP and Telegram OAuth authentication with shareable public conversations and SEO optimization (Open Graph, JSON-LD, SSR).
- Configured **CI/CD** with **GitHub Actions** and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Langfuse** for LLM observability, **Infisical** for secrets management, and **Playwright** E2E tests across 4 test projects.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*